



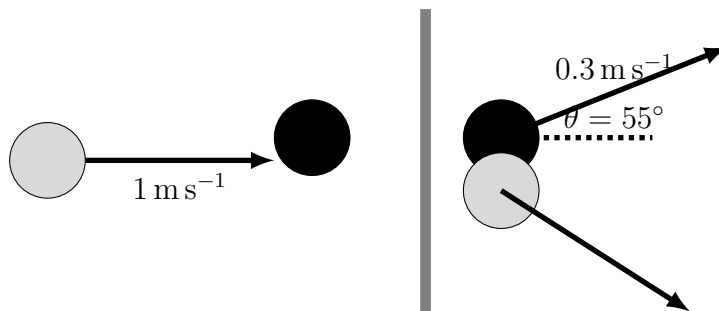
Physics Worksheet

Momentum and Centre of Mass

Name: _____ Student #: _____ Student Email #: _____

1. Imagine a game of pool on a peculiar pool table. The white cue is struck and strikes the side of the table where it compresses a spring, it then collides with the 8-ball and sends it into a pocket for the win. Below is a list of the forces acting on the **system of the two cue balls**. Explain whether these forces are external or internal. If they are external, explain what objects we would need to include in our system for them to be internal.
 - (a) Applied force on the white cue from the player
 - (b) Applied force of the collision between the two cue balls
 - (c) Gravity between the balls and The Earth
 - (d) The spring force from the spring
 - (e) The gravity between the two cue balls
 - (f) The friction between the table and the cue balls
2. A 1 m catapult is able to expend 15 kW of power over 0.1 s before launching its payload. What is the impulse on the payload from the catapult?

3. The gray 0.75 kg cue ball strikes the black 1.25 kg and they go flying off in different directions. At what velocity (magnitude and direction) does the gray cue ball go off in?



4. The following questions discuss centre of mass:
- (a) Does the centre of mass always reside *inside* of the object? What is an example of an object where this may not be the case
- (b) In what situations might it be more useful/easier to describe the motion of the centre of mass of an object as oppose to the object as a whole?
- (c) Describe a situation where object within a system are moving but the centre of mass of the system is stationary